PHASE 9

Business Insights & Recommendations

Turning technical analysis into executive decisions and an interview-ready business story

B2B Travel Booking & Revenue Analytics

Complete Understanding Notes | Beginner-Friendly | Portfolio & Interview Ready

Purpose: understand what was done, why it was done, how it was validated, and how to explain it in an interview.

Prepared from the completed project work and verified analysis results.

Contents and Learning Map

Use this document in two ways: first as a learning summary, then as an interview revision sheet. Each section explains the concept, implementation, validation, and business meaning.

#	Section	What you should understand
1	Executive summary	How to summarize business scale, profitability, growth, and risk.
2	Growth and revenue insights	What monthly volatility means and what to investigate.
3	Partner performance	Destination, supplier, hotel, and agent findings.
4	Cancellation and refund insights	The largest operational and financial cancellation drivers.
5	Prioritized recommendations	Actions ranked by business impact and investigation value.
6	Evidence vs causality	How to make recommendations without overclaiming.
7	Interview story	A concise end-to-end explanation of the complete project.

Project rule: Do not memorize code alone. Be able to explain the grain of the data, why each step is needed, and how you verified the result.

1. Executive Business Summary

Phase 9 answers the question that matters most to management: what should the business learn from the analysis, and what actions should be considered next?

Metric	Verified result	Business meaning
Analysis-ready bookings	119,877	Large enough to evaluate broad commercial patterns.
GBV	about INR 26.15B	Total booking value processed.
Net Revenue	about INR 3.25B	Revenue remaining after supplier cost and discount logic.
ABV	about INR 218.15K	Average booking value per valid booking.
Gross Margin	12.43%	Overall revenue efficiency relative to GBV.
Booking Cancellation Rate	about 11.64%	Meaningful booking leakage and operational risk.

Executive summary: The platform processed about 119.9K analysis-ready bookings worth INR 26.15B and generated about INR 3.25B in net revenue at a 12.43% gross margin. Performance varies by month and partner, while an approximately 11.64% booking cancellation rate creates a material operational and financial area for investigation.

2. Growth and Revenue Insights

Observation	Verified result	Interpretation
Strongest GBV MoM month	Dec 2024: +11.31%	Strong acceleration; investigate seasonality, mix, availability, and promotion context.
Weakest GBV MoM month	Feb 2024: -8.68%	Meaningful contraction; investigate demand and booking-mix changes.
Overall margin	12.43%	Growth should be evaluated together with net revenue and margin, not GBV alone.

Key lesson: Growth is not simply more bookings. Sustainable growth means stronger bookings/GBV together with healthy net revenue, margin, and controlled cancellations.

Recommended management view: track GBV growth, net-revenue growth, gross margin, and cancellation rate together. A month with high GBV but weak margin or elevated cancellations may not represent high-quality growth.

3. Commercial Partner Insights

Area	Verified leader / finding	Business interpretation
Destination	D0055 leads GBV and net revenue	Priority market for inventory availability and commercial support.
Supplier	S0079 leads GBV and net revenue	Strategically important supplier relationship.
Supplier margin	S0052 highest at 13.05%; S0143 lowest at 11.67%	Separate scale from margin efficiency; review weaker-margin economics.
Hotel GBV	H00978	Highest hotel booking-value contribution.
Hotel net revenue	H00516	Highest hotel net-revenue contribution.
Hotel margin	H02897 highest at 15.42%; H02297 lowest at 9.01%	Potential opportunity to compare supplier costs, discounting, and booking mix.
Agent	A00238 leads net revenue	High-value commercial partner worth retaining and supporting.
Agent bookings	A00145 leads with 249 bookings	Highest booking frequency, which is a different metric from revenue leadership.

Analyst wording: Avoid saying "best supplier" without naming the KPI. Top by GBV, top by net revenue, and top by margin are different business concepts.

4. Cancellation and Refund Insights

Finding	Verified result	Why it matters
Largest cancellation reason	Customer Change Of Plan - 4,823 events	Most frequent cancellation driver.
Largest total refund reason	Customer Change Of Plan - about INR 1.021B	Largest observed refund exposure by reason.
Highest average refund reason	Price Found Elsewhere - about INR 212.97K	Potential pricing-retention issue for higher-value bookings.

Finding	Verified result	Why it matters
Highest total cancellation fee reason	Customer Change Of Plan - about INR 36.32M	Also creates the largest fee amount in aggregate.

Business interpretation

Customer Change Of Plan is important because it is large on both frequency and total refund value. Price Found Elsewhere is important because the average refund is highest, suggesting that those cancellations can be financially significant even if they occur less frequently.

5. Prioritized Recommendations

Priority	Evidence	Recommended action	Metric to monitor
1	Customer Change Of Plan: 4,823 events and about INR 1.02B refunds	Test modification/rebooking alternatives before full cancellation.	Cancellation rate; refund value; modification acceptance
2	Price Found Elsewhere has highest average refund	Review rate competitiveness and supplier pricing for high-value bookings.	Price-found-elsewhere rate; retained booking value
3	D0055 leads GBV and net revenue	Protect hotel/supplier availability and monitor service quality in this destination.	GBV; net revenue; availability; cancellations
4	S0079 leads supplier contribution	Deepen commercial relationship while monitoring concentration risk.	Supplier GBV share; margin; cancellations
5	S0143 has lowest supplier margin among analyzed meaningful-volume suppliers	Review cost, discount, and contract economics.	Supplier margin; cost-to-GBV ratio
6	H02297 has lowest hotel margin in analyzed >=50 booking group	Investigate pricing, supplier cost, discounting, and booking mix.	Hotel margin; net revenue per booking
7	A00238 leads agent net revenue	Prioritize retention and partner support.	Agent net revenue; bookings; cancellation quality
8	MoM GBV varies from -8.68% to +11.31%	Include seasonality/volatility in forecasting and capacity planning.	Forecast error; monthly GBV growth

6. Evidence, Hypotheses, and Causal Discipline

Strong business analysis distinguishes between what the data proves and what should be investigated next. The project identifies patterns and commercial priorities; it does not automatically establish causal explanations.

Avoid saying	Better analyst language
"S0143 is a bad supplier."	"S0143 has the lowest observed margin among the analyzed supplier group, so cost, discount, and contract terms should be investigated."
"Price Found Elsewhere proves our prices are too high."	"Price Found Elsewhere has the highest average refund, which makes price competitiveness a priority hypothesis to test."
"December grew because of seasonality."	"December had the strongest MoM growth; seasonality is one plausible driver to investigate."
"High GBV outliers are wrong."	"High-value bookings appear associated with more rooms and longer stays, so many are plausible B2B transactions."

Professional rule: Finding -> business implication -> recommended action -> metric to monitor.

7. Complete Interview-Ready Business Story

A strong project explanation should connect the business problem, data engineering choices, analysis, dashboard, findings, and recommendations in one coherent story.

2-minute answer: I built an end-to-end B2B Travel Booking & Revenue Analytics project using PostgreSQL, Python/Pandas, Power BI, and Excel. The objective was to understand booking growth, revenue performance, profitability, and cancellation drivers. I worked with seven related datasets containing over 120,000 raw booking rows and realistic quality issues including duplicates, mixed dates, invalid booking values, missing suppliers, and multiple payments per booking. In PostgreSQL I created raw, staging, clean, and analytics layers, removed exact duplicate bookings, isolated invalid records, aligned payment grain, and normalized 15 currencies into INR. In Python I profiled booking distributions, investigated IQR outliers, analyzed monthly growth and partner performance, and studied cancellation/refund drivers. In Power BI I built a validated semantic model, DAX KPIs, time-intelligence measures, slicers, trend charts, top-N commercial views, and cancellation analysis. The final analysis contains 119,877 valid bookings, about INR 26.15B GBV, INR 3.25B net revenue, and a 12.43% gross margin. Dec 2024 had the strongest GBV MoM growth, D0055 and S0079 were major contributors, and Customer Change Of Plan was the largest cancellation/refund driver. I recommended cancellation-retention actions, pricing competitiveness reviews, stronger management of high-value partners, and investigation of weaker-margin suppliers and hotels.

Interview questions you should be ready for

Question	Core answer point
What was the main business problem?	Identify sources of growth, profitability, and cancellation leakage.
Why did you use multiple tools?	SQL for trusted data preparation, Python for flexible EDA, Power BI for interactive decision support.
What was the most important data-modeling decision?	Respecting grain, especially aggregating payments before joining to bookings.
What was your strongest insight?	Customer Change Of Plan dominates cancellation frequency and refund exposure; commercial performance is concentrated in specific partners.
What would you do next?	Test cancellation-retention interventions, investigate pricing competitiveness, and deepen segmentation by partner, booking value, and lead time.

8. Phase 9 Completion Checklist

- Executive summary connects KPIs to business meaning.
- Growth insights distinguish observation from cause.
- Destination, supplier, hotel, and agent findings are KPI-specific.
- Cancellation and refund findings are quantified.
- Recommendations are prioritized and measurable.
- Causal overclaims are avoided.
- A concise end-to-end interview story is ready.

Project outcome: The project now moves from raw data to a management-ready story: trustworthy data -> analysis -> dashboard -> insight -> recommended action.